

Chapter 10

The Euler equations

In this Chapter, we are concerned with presenting and discussing the unsteady Euler equations, which constitute a non-linear hyperbolic system of partial differential equations. We consider the one-dimensional, two-dimensional, and three-dimensional Euler equations. We discuss their so-called *dimensionally split* forms. Further, we explore modifications to the Cartesian form of the equations in order to accommodate cylindrical and spherical symmetries efficiently.

The material in this Chapter borrows heavily from Toro [2009] and the interested reader is encouraged to consult this excellent reference for more in-depth discussions.

10.1 One-dimensional Euler equations

The time-dependent, or unsteady Euler equations in one dimension are a system of non-linear hyperbolic conservation laws that govern the dynamics of a compressible fluid for which the effects of body forces, viscous stresses, and heat fluxes are neglected.

Despite their simplicity, the one-dimensional unsteady Euler equations are very versatile and exhibit the key behaviors of compressible fluids at elevated values of the Mach number, such as the formation of shocks, rarefaction waves, and contact waves. Thus, one may explore the performance of numerical methods on the one-dimensional Euler equations prior to considering the more complex three-dimensional, unsteady Euler equations.

Define the state vector $\mathbf{U} = \mathbf{U}(x, t)$ of conserved variables

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ E \end{bmatrix}, \quad (10.1)$$

where ρ is the density (SI units: kg m^{-3}), ρu is the x component of the fluid's momentum per unit volume (SI units: $\text{kg m}^{-2} \text{s}^{-1}$), and E is the *total energy* of the fluid (SI units: J m^{-3}) defined as

$$E = \rho \left(\frac{u^2}{2} + e \right), \quad (10.2)$$

with e and $u^2/2$ are the *specific internal energy* and *specific kinetic energy* of the fluid, respectively (SI units: J kg^{-1}).

Define the flux vector $\mathbf{F} = \mathbf{F}(\mathbf{U})$

$$\mathbf{F} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ u(E + p) \end{bmatrix}, \quad (10.3)$$

where p is pressure (SI units: N m^{-2}).

Then, the one-dimensional, unsteady Euler equations read

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = 0, \quad (10.4)$$

where the temporal and spatial derivatives are *element by element* or

$$\mathbf{U}_t + \mathbf{F}_x = 0 \quad (10.5)$$

Note that because the conservation laws feature the spatial derivative of a flux, they are in *conservative form*.

In order to close the system of equations, we consider an ideal gas so that the equation of state is

$$\rho R T = p, \quad (10.6)$$

where R is the gas constant (SI units: $\text{J kg}^{-1} \text{K}^{-1}$), T is temperature (SI units: K) and p is pressure. The gas constant is

$$R = \mathcal{R}/W, \quad (10.7)$$

where $\mathcal{R} = 8314 \text{ J kmol}^{-1} \text{ K}^{-1}$ is the universal gas constant and W is the molar mass of the gas (SI units: kg kmol^{-1}). Further, we assume that the gas is calorically perfect, so that

$$e = \frac{p}{\rho} \frac{1}{\gamma - 1} = \frac{R}{\gamma - 1} T = c_p T, \quad (10.8)$$

where $\gamma = c_p/c_v$ is the ratio of specific heats, $c_p = R/(\gamma - 1)$, and $c_v = c_p - R$. The speed of sound a is defined as

$$a = \sqrt{\gamma R T} = \sqrt{\gamma \frac{p}{\rho}}. \quad (10.9)$$

Note that Eq. (10.8) indicates that $p = p(\rho, e)$, so that one can recover p from the state vector $\mathbf{U}$

$$p = (\gamma - 1)\rho e = (\gamma - 1)(E - \rho u^2/2). \quad (10.10)$$

As an example, if we assume air has a molar mass of $W = 29 \text{ kg kmol}^{-1}$, $R = 8314/29 = 286.7 \text{ J kg}^{-1} \text{ K}^{-1}$. The ratio of specific heats of air is usually taken to be $\gamma = 1.4$.

10.1.1 Isothermal equations

Assuming that the temperature of the gas is constant (isothermal flow), only the conservation of mass and momentum are required. These are called the *isothermal equations* and read

$$\mathbf{U}_t + \mathbf{F}(\mathbf{U})_x = 0, \quad (10.11)$$

where

$$\mathbf{U} = \begin{vmatrix} \rho \\ \rho u \end{vmatrix} \quad (10.12)$$

and

$$\mathbf{F} = \begin{vmatrix} \rho u \\ \rho u^2 + \rho a^2 \end{vmatrix} \quad (10.13)$$

where $a = \sqrt{\gamma R T} = \text{const}$ is the constant speed of sound in the gas.

The isothermal equations are often used in the first phase of the development of numerical methods for the Euler equations.

10.1.2 Isentropic equations

In the limit of constant entropy (isentropic flow), only conservation of mass and momentum are required, since the pressure and density are related one to the other by the constraint that the entropy of the gas remains constant. These are called the *isentropic equations* and read

$$\mathbf{U}_t + \mathbf{F}(\mathbf{U})_x = 0, \quad (10.14)$$

where

$$\mathbf{U} = \begin{vmatrix} \rho \\ \rho u \end{vmatrix} \quad (10.15)$$

and

$$\mathbf{F} = \begin{vmatrix} \rho u \\ \rho u^2 + p \end{vmatrix} \quad (10.16)$$

where $p = C\rho^\gamma$ provides a closure condition ($C = \text{const}$) consistent with the specific entropy of the gas remaining constant.

10.2 The Jacobian matrix of the one-dimensional Euler equations

The one-dimensional Euler equations feature a flux vector $\mathbf{F}$, which is a function of the state vector of the conservative variables, i.e. $\mathbf{F}(\mathbf{U})$. The *Jacobian matrix* or simply *Jacobian* is a matrix defined as

$$\mathbf{A}(\mathbf{U}) = \frac{\partial \mathbf{F}(\mathbf{U})}{\partial \mathbf{U}}, \quad (10.17)$$

i.e. each element of the Jacobian matrix is a partial derivative of selected flux components with respect to a component of the state vector: $(A)_{ij} = \partial F_i / \partial U_j$.

Before we can write explicitly the Jacobian matrix, it is useful to rearrange the flux vector so that the dependence of the flux vector from the state vector of the conservative variables is explicit. Start by recalling that $\mathbf{U} = (u_1, u_2, u_3)$ where

$$u_1 = \rho \quad (10.18)$$

$$u_2 = \rho u \quad (10.19)$$

$$u_3 = E \quad (10.20)$$

and $\mathbf{F} = (f_1, f_2, f_3)$ where

$$f_1 = \rho u \quad (10.21)$$

$$f_2 = \rho u^2 + p \quad (10.22)$$

$$f_3 = u(E + p). \quad (10.23)$$

Now manipulate each element of the flux vector individually

$$f_1 = \rho u = u_2 \quad (10.24)$$

$$\begin{aligned} f_2 &= \rho u^2 + p \\ &= \rho u^2 + (\gamma - 1)(E - \rho u^2/2) \\ &= (\rho u)^2/\rho + (\gamma - 1)[u_3 - (\rho u)^2/(2\rho)] \\ &= u_2^2/u_1 + (\gamma - 1)[u_3 - (u_2^2)/(2u_1)] \\ &= (3 - \gamma)u_2^2/(2u_1) + (\gamma - 1)u_3 \end{aligned} \quad (10.25)$$

$$\begin{aligned} f_3 &= u(E + p) = uE + up \\ &= uE + u(\gamma - 1)(E - \rho u^2/2) \\ &= \gamma uE - (\gamma - 1)\rho u^3/2 \\ &= \gamma u_2 u_3/u_1 - (\gamma - 1)u_2^3/(2u_1^2). \end{aligned} \quad (10.26)$$

In summary, we have manipulated the expressions of the elements of the flux vector to expose the dependence from the elements of the state vector of conservative variables

$$\mathbf{F} = \mathbf{F}(\mathbf{U}) = \begin{pmatrix} f_1 \\ f_2 \\ f_3 \end{pmatrix} = \begin{pmatrix} u_2 \\ (3 - \gamma)u_2^2/(2u_1) + (\gamma - 1)u_3 \\ \gamma u_2 u_3/u_1 - (\gamma - 1)u_2^3/(2u_1^2) \end{pmatrix}. \quad (10.27)$$

Next, it is straightforward, albeit tedious, to obtain the expression for the Jacobian matrix, which requires to take partial derivatives $(A)_{ij} = \partial f_i / \partial u_j$. One obtains

$$\mathbf{A}(\mathbf{U}) = \begin{pmatrix} 0 & 1 & 0 \\ (\gamma - 3)u^2/2 & (3 - \gamma)u & \gamma - 1 \\ (\gamma - 2)u^3/2 - a^2 u/(\gamma - 1) & (3 - 2\gamma)u^2/2 + a^2/(\gamma - 1) & \gamma u \end{pmatrix}. \quad (10.28)$$

The system of equations is hyperbolic with the following real eigenvalues

$$\lambda_1 = u - a, \quad (10.29)$$

$$\lambda_2 = u, \quad (10.30)$$

$$\lambda_3 = u + a. \quad (10.31)$$

The matrix of the corresponding right eigenvectors is

$$\mathbf{K} = \left[\mathbf{K}^{(1)} \quad \mathbf{K}^{(2)} \quad \mathbf{K}^{(3)} \right] = \begin{bmatrix} 1 & 1 & 1 \\ u - a & u & u + a \\ H - ua & u^2/2 & H + ua \end{bmatrix}, \quad (10.32)$$

where $H = (E + p)/\rho = u^2/2 + h$ is the *total specific enthalpy* (SI units: J kg⁻¹) and h is the *specific enthalpy* (SI units: J kg⁻¹).

Notice that the eigenvalues of the Jacobian are related to the local fluid speed u and the local speed of sound a . The sign of the eigenvalues is related to whether the fluid is left moving, right moving and to the local Mach number $\text{Ma} = |u|/a$. In other words, depending on whether the speed of the fluid, as described by the state vector of the conserved variables, is supersonic or subsonic and which direction the fluid is moving, different scenarios are possible. These different scenarios have very important implications for the numerical solution of the Euler equations as we shall discuss in the remainder of the Chapter.

A second, important observation pertains to the so-called *homogeneity property* of the Euler equations, which we explore below.

10.2.1 Homogeneity property of the Euler equations

The one-dimensional Euler equations shown in Section 10.1 satisfy the homogeneity property

$$\mathbf{F}(\mathbf{U}) = \mathbf{A}(\mathbf{U})\mathbf{U}, \quad (10.33)$$

where $\mathbf{A}(\mathbf{U}) = \partial \mathbf{F} / \partial \mathbf{U}$ is the Jacobian matrix discussed in Section 10.2. In other words, the flux vector is equal to the product of the Jacobian matrix and the state vector.

The proof of the above statement is straightforward once the Jacobian matrix is written in terms of the components of the state vector of the con-

served variables as

$$\mathbf{A}(\mathbf{U}) = \begin{vmatrix} 0 & 1 & 0 \\ (3-\gamma)(u_2/u_1)^2/2 & (3-\gamma)u_2/u_1 & \gamma-1 \\ -\gamma u_2 u_3/u_1^2 + (\gamma-1)(u_2/u_1)^3 & \gamma u_3/u_1 - 3(\gamma-1)(u_2/u_1)^2/2 & \gamma u_2/u_1 \end{vmatrix}. \quad (10.34)$$

The homogeneity property of the Euler equations is rather useful in the derivation of flux vector splitting methods as explained in Chapter ???. The property holds also for the two- and three-dimensional Euler equations, although the proof requires more manipulation on the account of the additional dimensions and terms.

The homogeneity property holds for the isothermal and isentropic equations also. The proof is left to the reader.

10.3 Two-dimensional Euler equations

The two-dimensional Euler equations are very similar to the one-dimensional ones, but feature one additional component of momentum, one additional equation, and derivatives in the second dimension y .

Define the state vector $\mathbf{U} = \mathbf{U}(x, y, t)$ of conserved variables

$$\mathbf{U} = \begin{vmatrix} \rho \\ \rho u \\ \rho v \\ E \end{vmatrix}, \quad (10.35)$$

where ρ is the density (SI units: kg m^{-3}), ρu and ρv are the x and y components of the fluid's momentum per unit volume (SI units: $\text{kg m}^{-2} \text{s}^{-1}$), and E is the total energy per unit volume (SI units: J m^{-3}) defined as

$$E = \rho \left(\frac{|\mathbf{u}|^2}{2} + e \right), \quad (10.36)$$

with e and $|\mathbf{u}|^2/2 = (u^2 + v^2)/2$ are the internal energy and the kinetic energy per unit mass of the fluid, respectively (SI units: J kg^{-1}).

Define the flux vector $\mathbf{F} = \mathbf{F}(\mathbf{U})$ in the x direction

$$\mathbf{F} = \begin{vmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(E + p) \end{vmatrix}, \quad (10.37)$$

where p is pressure (SI units: N m^{-2}). Similarly, define the flux vector $\mathbf{G} = \mathbf{G}(\mathbf{U})$ in the y direction

$$\mathbf{G} = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(E + p) \end{bmatrix}, \quad (10.38)$$

The equations are complemented by the equation for pressure

$$p = (\gamma - 1)\rho e = (\gamma - 1)(E - \rho|\mathbf{u}|^2/2). \quad (10.39)$$

Then, the two-dimensional, unsteady Euler equations read

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} = 0, \quad (10.40)$$

where the temporal and spatial derivatives are *element by element* or

$$\mathbf{U}_t + \mathbf{F}_x + \mathbf{G}_y = 0 \quad (10.41)$$

Note that because the conservation laws feature the spatial derivative of fluxes, they are in *conservative form*.

10.4 Three-dimensional Euler equations

The three-dimensional Euler equations are very similar to the two-dimensional ones, but feature one additional component of momentum, one additional equation, and derivatives in the third dimension z .

Define the state vector $\mathbf{U} = \mathbf{U}(x, y, z, t)$ of conserved variables

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ E \end{bmatrix}, \quad (10.42)$$

where ρ is the density (SI units: kg m^{-3}), ρu , ρv , and ρw are the x , y , and z components of the fluid's momentum per unit volume (SI units: $\text{kg m}^{-2} \text{s}^{-1}$), and E is the total energy per unit volume (SI units: J m^{-3}) defined as

$$E = \rho \left(\frac{|\mathbf{u}|^2}{2} + e \right), \quad (10.43)$$

with e and $|\mathbf{u}|^2/2 = (u^2 + v^2 + w^2)/2$ are the internal energy and the kinetic energy per unit mass of the fluid, respectively (SI units: J kg⁻¹).

Define the flux vector $\mathbf{F} = \mathbf{F}(\mathbf{U})$ in the x direction

$$\mathbf{F} = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho uw \\ u(E + p) \end{pmatrix}, \quad (10.44)$$

where p is pressure (SI units: N m⁻²). Similarly, define the flux vector $\mathbf{G} = \mathbf{G}(\mathbf{U})$ in the y direction

$$\mathbf{G} = \begin{pmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ \rho vw \\ v(E + p) \end{pmatrix}. \quad (10.45)$$

Finally, define the flux vector $\mathbf{H} = \mathbf{H}(\mathbf{U})$ in the z direction

$$\mathbf{H} = \begin{pmatrix} \rho w \\ \rho uw \\ \rho vw \\ \rho w^2 + p \\ w(E + p) \end{pmatrix}. \quad (10.46)$$

Then, the three-dimensional, unsteady Euler equations read

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} + \frac{\partial \mathbf{H}}{\partial z} = 0, \quad (10.47)$$

where the temporal and spatial derivatives are *element by element* or

$$\mathbf{U}_t + \mathbf{F}_x + \mathbf{G}_y + \mathbf{H}_z = 0 \quad (10.48)$$

Note that because the conservation laws feature the spatial derivative of fluxes, they are in *conservative form*.

Bibliography

E F Toro. *Riemann solvers and numerical methods for fluid dynamics*. Springer-Verlag, Berlin Heidelberg, 3 edition, 2009.

